
Software Developer (Java Developer – Scale I)

Syllabus

1. Core Java Programming

- Java fundamentals (OOP concepts, classes, objects, inheritance, polymorphism)
 - Exception handling
 - Collections Framework (List, Set, Map)
 - Multithreading and concurrency
 - File handling and serialization
 - Java memory management and JVM basics
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2. Backend Development (Java Frameworks)

- **Spring Framework & Spring Boot**
 - Dependency Injection (DI) / IoC
 - RESTful API development
 - Spring MVC architecture
 - Spring Data / JPA basics
 - **Reactive Programming concepts**
 - Backend service development using Java
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3. Frontend Development

- HTML and CSS
- **JavaScript fundamentals**
- Frontend frameworks
 - **React JS**
 - **Angular**
- **Node.js and Express.js**

- Client-server interaction and UI integration with APIs
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4. API Development & Integration

- **REST API design principles**
 - **SOAP-based web services**
 - API request/response structure
 - API security basics
 - JSON, XML and delimited data formats
 - API lifecycle management
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5. Enterprise Integration (IBM ACE / MQ)

- **IBM App Connect Enterprise (ACE)**
 - Integration server configuration
 - Message flows and integration interfaces
 - BAR file deployment
 - **Java Compute Node (JCN) framework**
 - **ESQL programming**
 - Message transformation and routing
 - Message models using **XML, MRM, DFDL**
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6. Messaging Systems

- **IBM MQ architecture**
 - MQ channels and transmission protocols
 - MQ client applications
 - Queue managers and message queues
 - Messaging patterns and asynchronous communication
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7. Microservices Architecture

- Microservices design principles
- Service communication patterns
- Service discovery and API gateways

- Fault tolerance and scalability
 - Container-based microservice deployment
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8. Containers & DevOps

- **Containerization (Docker basics)**
 - Application deployment in containers
 - **CI/CD pipeline concepts**
 - Automated build and deployment
 - Version control and release management
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9. API Management Platforms

- **IBM API Connect**
 - API Manager
 - Developer Portal
 - Developer Toolkit
 - API publishing and governance
 - API lifecycle management
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10. Software Development Lifecycle

- Coding standards and best practices
 - Debugging and troubleshooting
 - Application maintenance and support
 - **User Acceptance Testing (UAT)**
 - Information security testing
 - Documentation and knowledge sharing
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11. Production Support & Monitoring

- Handling production bugs
- Monitoring integration servers
- Resource management of integration nodes
- Deployment management and system health monitoring

12. Data Handling & Error Management

- Data analysis techniques
 - Scalable application design
 - Error handling and logging strategies
 - High-availability application architecture
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Mobile Developer- Android/IOS

1. Programming Fundamentals

- **Object-Oriented Programming (OOP)** – classes, inheritance, polymorphism, abstraction
 - **SOLID principles and design patterns**
 - Programming languages
 - **Java**
 - **Kotlin**
 - **Swift**
 - Basic **Objective-C interoperability with Swift**
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2. Android Application Development

- **Android SDK and Android Studio**
- Android application components
 - Activities
 - Fragments
 - Services
 - Broadcast Receivers
- UI development using **XML layouts and Material Design**
- **Android Architecture Patterns**
 - MVC

- MVP
 - MVVM
 - Data persistence
 - **SQLite**
 - **Room Database**
 - Threading and background processing
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3. iOS Application Development

- **Swift programming language**
 - Development using **Xcode**
 - iOS frameworks
 - UIKit
 - SwiftUI
 - CoreData
 - App lifecycle management
 - Memory management and performance optimization
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4. Cross-Platform Mobile Development

- **React Native**
 - **Flutter**
 - Hybrid mobile frameworks
 - Cross-platform UI components and architecture
 - Integration with native mobile features
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5. Mobile UI / UX Development

- Mobile UI design principles
 - **Material Design guidelines**
 - Custom UI component development
 - Responsive UI for multiple device sizes
 - Performance optimization and smooth UI rendering
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6. API Integration & Networking

- **RESTful API development and consumption**
 - **SOAP-based web services**
 - Data exchange formats
 - JSON
 - XML
 - Networking libraries and API communication
 - **GraphQL basics**
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7. Data Management & Offline Storage

- Local data persistence
 - SQLite
 - Room
 - CoreData
 - Offline data synchronization
 - Caching strategies
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8. Mobile Backend Integration

- Backend service integration
 - Secure API communication
 - Authentication and authorization
 - Third-party SDK integration
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9. Cloud Messaging & Notifications

- Push notification services
 - Cloud messaging APIs
 - Background notifications and event handling
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10. Functional Reactive Programming

- Reactive programming concepts
- Frameworks

- **RxSwift**
 - **Combine**
-

11. Mobile App Performance Optimization

- Memory management
 - Battery optimization
 - App performance monitoring
 - Threading and asynchronous programming
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12. Mobile App Security

- Secure data storage
 - API security
 - Authentication mechanisms
 - Secure coding practices
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13. Mobile App Lifecycle & Deployment

- App lifecycle stages
 - Concept
 - Design
 - Development
 - Testing
 - Deployment
 - Maintenance
 - Application packaging and release
 - **Publishing apps to Google Play Store and Apple App Store**
 - Version control and release cycles
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14. Testing & Debugging

- Unit testing
- UI testing
- Debugging mobile applications

- Performance testing and bug fixing

15. Mobile Development Ecosystem

- Third-party libraries and SDKs
 - Mobile architecture trends
 - Emerging mobile technologies
 - Collaboration with backend and product teams
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Software Developer – .NET

1. Programming Fundamentals

- **C# programming language**
 - Object-Oriented Programming (OOP)
 - Encapsulation
 - Inheritance
 - Polymorphism
 - Abstraction
 - Data structures and algorithms
 - Exception handling
 - Multithreading and asynchronous programming
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2. .NET Framework & .NET Core

- Architecture of **.NET Framework / .NET Core**
- **ASP.NET application development**
- **ASP.NET MVC architecture**
- Web application development using **Visual Studio**
- Middleware and dependency injection

- Configuration and environment management
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3. Web Development Technologies

- **HTML5**
 - **CSS / Bootstrap**
 - JavaScript fundamentals
 - Client-side and server-side interaction
 - Responsive web application design
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4. Backend Development

- Developing backend services using **C# and .NET**
 - Server-side business logic implementation
 - Integration of frontend components with backend services
 - API development and backend service architecture
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5. API Development & Integration

- **RESTful API development**
 - Web services integration
 - API design and documentation
 - Secure API communication
 - Integration with external services and applications
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6. Software Architecture & Design

- Software architecture principles
 - **MVC architecture**
 - Design patterns and best practices
 - System architecture design
 - Scalable and distributed system design
 - Enterprise software architecture
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7. Microservices Architecture

- Microservices concepts and architecture
 - Service communication patterns
 - Distributed systems design
 - Service scalability and reliability
 - Integration between microservices
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8. Database Design & Data Management

- Database design principles
 - Relational database concepts
 - Data modeling and schema design
 - Data access and data persistence techniques
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9. DevOps & CI/CD

- **Continuous Integration / Continuous Deployment**
 - Automated build and deployment pipelines
 - Version control systems (**Git**)
 - DevOps practices in software development lifecycle
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10. Software Development Lifecycle (SDLC)

- Business Requirement Analysis
 - System design and implementation
 - Coding standards and best practices
 - Testing phases
 - Unit testing
 - User Acceptance Testing (UAT)
 - Security testing
 - Software deployment and maintenance
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11. Application Security

- Secure coding practices
- Data protection mechanisms

- Authentication and authorization
 - Web application security concepts
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12. Agile Development & Project Management

- Agile software development methodology
 - Scrum concepts
 - Managing development teams
 - Project planning and delivery
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13. Performance Optimization & Debugging

- Application performance tuning
 - Troubleshooting and debugging
 - Scalability optimization
 - High availability system design
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14. Enterprise Application Development

- Building enterprise-level applications
 - Cross-functional system integration
 - Scalable architecture implementation
 - Software documentation and technical reporting
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Database Administration (DBA)

1. Database Fundamentals

- Database management system concepts (DBMS)
- Relational vs NoSQL databases
- Data models and schema design
- Data normalization and indexing techniques

- Query processing and optimization
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2. SQL & Database Programming

- **SQL queries** (DDL, DML, DCL, TCL)
 - Joins, subqueries, views, stored procedures
 - Query optimization techniques
 - Database scripting using **SQL, Python, Perl**
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3. Oracle Database Administration

- **Oracle database architecture**
 - Instance, memory structures, background processes
 - Oracle installation, configuration and maintenance
 - **Oracle Enterprise Manager (OEM)** for monitoring
 - Database upgrades and migrations
 - Oracle database features (Partitioning, OLAP, Spatial)
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4. PostgreSQL Administration

- PostgreSQL architecture and configuration
 - Database installation and setup
 - Performance monitoring and tuning
 - User management and privilege control
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5. NoSQL Databases

- **MongoDB architecture and administration**
 - **Redis in-memory database concepts**
 - Data storage models in NoSQL systems
 - Use cases for NoSQL databases
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6. Database Performance Tuning

- Query performance optimization
- Index optimization

- Monitoring database workload
 - Disk, memory and I/O management
 - Identifying and resolving performance bottlenecks
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7. Backup, Recovery & Disaster Management

- Database backup strategies
 - **Oracle RMAN backup and recovery**
 - Point-in-time recovery
 - Disaster recovery planning
 - Business continuity planning
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8. High Availability & Replication

- Database replication concepts
 - **Oracle Data Guard / Active Data Guard**
 - **Oracle Real Application Clusters (RAC)**
 - Data redundancy and failover mechanisms
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9. Database Security

- User account management
 - Role and privilege management
 - Principle of least privilege
 - Encryption and secure database configuration
 - Secure network communication (SSL/TLS)
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10. Database Auditing & Compliance

- Database audit policies and logging
 - Monitoring security incidents
 - Security event reporting and alerts
 - Compliance with regulatory and security standards
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11. Data Protection & Governance

- Data masking techniques
 - Access control policies
 - Secure storage of sensitive data
 - Data integrity and protection mechanisms
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12. Database Monitoring & Automation

- Monitoring tools (Oracle Enterprise Manager)
 - Log monitoring and alert systems
 - Automation and scripting for database administration
 - Operational efficiency and task automation
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13. Data Integration & Replication Tools

- **Oracle GoldenGate / CDC (Change Data Capture)**
 - Data replication techniques
 - Data synchronization across systems
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14. Capacity Planning & Infrastructure

- Database storage management
 - Capacity planning and scaling
 - Hardware resource utilization
 - Infrastructure optimization
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High-Priority

- SQL queries and database design
- Oracle database architecture
- PostgreSQL administration
- MongoDB and Redis basics
- Database performance tuning
- Oracle RMAN backup & recovery
- Oracle RAC and Data Guard
- Database security and user privilege management

- GoldenGate / CDC replication
 - Storage, memory and I/O optimization
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Digital Channel Manager / Digital Application Specialist

1. Digital Banking Systems

- Digital banking platforms and architecture
- **Internet Banking and Mobile Banking systems**
- Digital channel infrastructure and application servers
- Digital transaction processing systems

Digital payment systems

- **UPI**
 - **IMPS**
 - **NEFT**
 - **RTGS**
 - **AEPS**
 - **ATM Switch**
 - **POS / QR payments**
 - **Cards ecosystem**
 - **CBDC basics**
 - **ASBA / Demat systems**
 - **SWIFT messaging system**
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2. Payment Systems & Banking Infrastructure

- Payment gateway architecture
- NPCI payment ecosystem
- Card payment networks and transaction flow
- Settlement and clearing mechanisms

- Digital transaction lifecycle
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3. API Integration & Middleware

- **RESTful API architecture**
 - API lifecycle management
 - Middleware and integration frameworks
 - Microservices architecture
 - Third-party system integration
 - Payment gateway integrations
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4. Cloud & DevOps Technologies

- Cloud computing fundamentals
 - **AWS**
 - **Microsoft Azure**
 - **Google Cloud Platform**

DevOps practices

- CI/CD pipelines
 - Containerization (**Docker**)
 - Container orchestration (**Kubernetes**)
 - Application deployment automation
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5. Databases & Application Infrastructure

- Database concepts and management
 - **Oracle / SQL databases**
 - **NoSQL databases**
 - Application server architecture
 - Data storage and retrieval optimization
-

6. Cybersecurity & Fraud Prevention

- Cybersecurity frameworks and best practices
- Encryption techniques and secure communications

- Identity and access management
 - Fraud detection and prevention mechanisms
 - Secure digital transaction processing
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7. Incident Management & System Monitoring

- Monitoring tools and system health checks
 - Incident management processes
 - Root cause analysis (RCA)
 - Service availability and uptime management
 - Disaster recovery and business continuity planning
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8. Regulatory Compliance & Governance

- Banking regulatory guidelines
 - **RBI regulations**
 - **NPCI standards**
 - **PCI-DSS compliance**
 - Audit requirements for digital banking systems
 - Data privacy and security compliance
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9. Digital Banking Innovations

- **Artificial Intelligence in banking**
 - Chatbots and virtual assistants
 - Biometric authentication
 - Cloud migration strategies
 - Fintech ecosystem and emerging technologies
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10. Retail Banking & Payment Ecosystem

- Retail banking products and services
- Digital transaction trends
- Customer experience optimization
- Fintech integrations and innovations

11. Vendor & Technology Management

- Vendor management and SLA monitoring
- RFP process and technology evaluation
- Technology contract management
- Collaboration with IT vendors and service providers

12. IT Project & Team Management

- Translating business requirements into technical solutions
- Cross-functional team collaboration
- IT project lifecycle management
- Leadership and stakeholder management

High-Priority Topics for Exam

- UPI, IMPS, NEFT, RTGS architecture
- Digital banking systems (Mobile / Internet Banking)
- API integration and microservices
- Cloud platforms (AWS / Azure / GCP)
- DevOps tools (Docker, Kubernetes, CI/CD)
- Cybersecurity and fraud prevention in digital banking
- RBI / NPCI / PCI-DSS compliance
- Disaster recovery and incident management
- Payment gateway and card ecosystem
- Fintech innovations and digital banking trends

Digital Applications Specialist

1. Software Application Support & Maintenance

- Application support concepts and responsibilities

- Production support for enterprise applications
 - Application performance monitoring and optimization
 - Issue resolution and troubleshooting techniques
 - Minimizing downtime and service disruptions
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2. Programming & Application Development

- **Java / J2EE fundamentals**
- MVC frameworks and **Spring / Spring Boot**
- Service Oriented Architecture (SOA)
- **RESTful web services**

Additional programming exposure

- **NodeJS**
 - **Python**
 - **PHP**
 - **Go**
 - **Ruby**
 - **.NET / C#**
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3. Database Management

- Relational databases
 - **Oracle**
 - **PostgreSQL**
 - **MySQL**
- NoSQL databases
 - **MongoDB**

Database topics

- SQL queries and PL/SQL basics
 - Database troubleshooting
 - Basic **DBA tasks and database monitoring**
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4. Banking Application Systems

- Core Banking Systems (**CBS**)
 - Loan Origination Systems (**LOS**)
 - Mobile banking applications
 - Enterprise Service Bus (**ESB**)
 - API Gateway architecture
-

5. Application Monitoring & Troubleshooting

- Application monitoring tools
 - Log analysis and debugging techniques
 - Audit trail monitoring
 - Incident detection and resolution
 - Performance issue diagnosis
-

6. Software Testing

- **End-to-End application testing**
 - Functional testing
 - Bug identification and resolution
 - Software updates and patch implementation
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7. Agile Development & Deployment

- Agile methodology concepts
 - Software development lifecycle (SDLC)
 - Continuous deployment of applications
 - Agile team collaboration and release cycles
-

8. IT Service Management (ITSM)

- ITSM tools for issue tracking
 - Incident, problem and change management
 - Ticket triaging and resolution processes
 - **ITIL framework basics**
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9. Security & Compliance

- Application security concepts
 - **VAPT (Vulnerability Assessment & Penetration Testing)** fixes
 - Secure application configuration
 - Regulatory compliance for banking applications
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10. Disaster Recovery & High Availability

- Disaster Recovery (DR) planning
 - **DR drills and geo-site replication**
 - Application and database redundancy
 - Business continuity planning
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11. Reporting & Data Analysis

- Extracting customized reports from applications
 - Data analysis for troubleshooting
 - Monitoring usage and performance metrics
-

12. Technical Communication & Coordination

- Working with **L1 / L2 / L3 support teams**
 - Vendor coordination and support management
 - Understanding business requirements
 - Documentation and knowledge base creation
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Linux / Solaris / SAN / Storage / SRE / OpenShift

Linux / UNIX / Solaris Administrator / Site Reliability Engineer – Scale II / III

1. Linux / UNIX / Solaris Administration

- **Red Hat Enterprise Linux (RHEL) administration**

- Solaris operating system architecture
 - Linux/UNIX commands and system utilities
 - System installation, configuration, and maintenance
 - User and group management
 - Access control and credential management
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2. Shell Scripting & Automation

- Shell scripting using
 - **Bash**
 - **sh**
 - **ksh**
 - Automation of system administration tasks
 - Job scheduling and script execution
 - Log parsing and system monitoring automation
-

3. System Monitoring & Observability

- System monitoring tools
 - Observability stack concepts
 - Log management and log analysis
 - Performance monitoring and resource utilization tracking
 - Alerting and incident detection
-

4. Linux Package & Patch Management

- Software installation and updates
 - **YUM / RPM package management**
 - Dependency resolution
 - Patch management and system updates
 - Offline repository creation and maintenance
-

5. Storage Administration

- Storage architecture and management

- Storage types
 - **SAN (Storage Area Network)**
 - **NAS (Network Attached Storage)**
 - **DAS (Direct Attached Storage)**
 - Storage provisioning and allocation
 - Storage performance monitoring
 - Capacity planning and optimization
-

6. Backup & Disaster Recovery

- Backup strategies and policies
 - Backup tools (rsync, tar)
 - Automated backup scheduling
 - Data restoration procedures
 - Disaster recovery planning
 - Business continuity planning
-

7. High Availability & Clustering

- High availability (HA) concepts
 - Solaris clusters and Linux clusters
 - Failover and load balancing
 - Disaster recovery orchestration
 - Replication mechanisms
-

8. Virtualization & Infrastructure

- Virtualization concepts
- Solaris Zones and SPARC LDOMs
- Hardware architectures
 - x86
 - ppc64
 - sparc64
 - s390x

- Infrastructure provisioning and management
-

9. Storage Security & Data Protection

- Storage access control
 - Data encryption techniques
 - Secure storage configurations
 - Data protection mechanisms
 - Security hardening of servers
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10. Storage Technologies & Optimization

- Thin provisioning
 - Data deduplication
 - Compression techniques
 - Disk-to-disk storage systems
 - Storage performance optimization
-

11. Storage Hardware & Vendor Systems

- Storage solutions from vendors
 - **Dell EMC**
 - **HPE**
 - **IBM**
 - **Hitachi**
 - SAN switches and storage controllers
 - Tape libraries and backup hardware
-

12. Network Configuration & Troubleshooting

- IP addressing and routing
 - Network configuration in Linux
 - Troubleshooting network connectivity issues
 - Secure communication between servers
-

13. DevOps & Site Reliability Engineering

- SRE principles and practices
 - Infrastructure automation
 - System reliability and scalability
 - Collaboration with development and operations teams
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14. Security & System Hardening

- Linux security policies
 - Server hardening techniques
 - Security testing and vulnerability checks
 - Identity and access management integration
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15. Documentation & Change Management

- System documentation practices
 - Change management processes
 - Incident reporting and analysis
 - Collaboration with cross-functional IT teams
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Windows & Active Directory (AD) Administration

1. Windows Server Administration

- Windows Server architecture and features
 - **Windows Server 2012 / 2016 / 2019 / 2022 administration**
 - Installation, configuration and maintenance of Windows servers
 - Server roles and features management
 - System monitoring and troubleshooting
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2. Active Directory Infrastructure

- **Active Directory architecture**

- AD Forests and Domains
 - Domain Controllers configuration
 - Organizational Units (OU)
 - Trust relationships between domains
 - Active Directory replication
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3. Group Policy Management

- **Group Policy Objects (GPO)**
 - Policy creation and configuration
 - Security policy enforcement
 - GPO troubleshooting
 - Endpoint and server policy management
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4. Identity & Access Management

- User and group account management
 - Authentication and authorization mechanisms
 - **LDAPS secure directory access**
 - Pass-through authentication services
 - Identity federation concepts
-

5. Active Directory Services

- **Active Directory Federation Services (ADFS)**
 - Directory synchronization
 - Integration of applications with Active Directory
 - Identity management solutions
-

6. DNS & Directory Integration

- Internal DNS configuration and management
- DNS records and name resolution
- DNS integration with Active Directory
- DNS troubleshooting

7. Microsoft Certificate Services

- **Microsoft Certificate Authority (CA)**
- Certificate templates and management
- Public Key Infrastructure (PKI) concepts
- Digital certificates and encryption

8. Patch Management

- Windows patch management concepts
- Patch deployment and monitoring
- Patch testing and validation
- System updates and compliance management

9. Exchange & Email Infrastructure

- **Microsoft Exchange Server architecture**
- Exchange Server administration
- Integration with Active Directory
- **Office 365 / Exchange Online integration**

10. Automation & Scripting

- **PowerShell scripting**
- Automation of administrative tasks
- System configuration scripts
- Active Directory automation

11. Security & Compliance

- Windows server security policies
 - Secure authentication methods
 - Access control and privilege management
 - Security monitoring and compliance enforcement
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12. Troubleshooting & Support

- Active Directory troubleshooting techniques
 - DNS and authentication troubleshooting
 - Performance monitoring and diagnostics
 - Support for L1 / L2 / L3 teams
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13. System Integration & Infrastructure Management

- Application integration with Active Directory
 - Infrastructure planning and design
 - Directory services architecture management
 - High availability and scalability planning
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Cloud Manager

1. Cloud Computing Fundamentals

- Cloud computing architecture and service models
 - **IaaS, PaaS, SaaS**
 - Public, private and **multi-cloud environments**
 - Cloud deployment models and service delivery
 - Software Defined Data Center (SDDC)
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2. Virtualization Technologies

- Virtualization concepts and architecture
 - **Hyper-V virtualization platform**
 - **VMware virtualization platform**
 - Virtual machine provisioning and management
 - Virtualized infrastructure administration
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3. Data Center Infrastructure Management

- Software defined data center architecture
 - Infrastructure provisioning and configuration
 - Hardware and software deployment
 - High availability and scalability design
 - Infrastructure performance monitoring
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4. Linux / Unix System Administration

- Linux operating systems
 - **RHEL**
 - **Ubuntu**
 - Linux system configuration and management
 - User management and permissions
 - System monitoring and troubleshooting
-

5. Windows Server Administration

- Windows Server architecture
 - **Windows Server 2012 / 2016 / 2022 administration**
 - Server configuration and maintenance
 - Patch management and system updates
-

6. Active Directory Infrastructure

- **Active Directory architecture**
 - AD forests, domains and domain controllers
 - Organizational Units (OU) and trust relationships
 - **Group Policy Objects (GPO)** configuration
 - Active Directory integration with applications
-

7. Identity & Access Management

- Authentication and authorization mechanisms
- **Active Directory Federated Services (ADFS)**
- Pass-through authentication services

- **LDAPS secure directory access**
-

8. DNS & Directory Services

- DNS architecture and configuration
 - DNS integration with Active Directory
 - Domain name resolution and troubleshooting
-

9. Microsoft Certificate Services

- **Public Key Infrastructure (PKI)** concepts
 - Microsoft Certificate Authority administration
 - Certificate templates and management
 - Secure communication using digital certificates
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10. Patch Management

- Patch management lifecycle
 - Patch deployment and monitoring
 - Patch testing and validation
 - Windows system patching and compliance
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11. Email Infrastructure Integration

- **Microsoft Exchange Server architecture**
 - Exchange server administration
 - Integration with Active Directory
 - **Office 365 / Exchange Online integration**
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12. Security & Compliance

- Cloud security principles
 - Infrastructure security design
 - Access control and authentication
 - Compliance and governance policies
-

13. Automation & Infrastructure Operations

- Infrastructure automation concepts
 - PowerShell scripting for automation
 - System monitoring and troubleshooting
 - Incident resolution and operational management
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14. Collaboration & Infrastructure Management

- Cross-functional IT team coordination
 - Application integration with infrastructure
 - Infrastructure planning and architecture design
 - Technical consulting and advisory roles
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High-Priority Topics for Exam

- Cloud computing models and architecture
 - Virtualization (**VMware / Hyper-V**)
 - Linux server administration (RHEL / Ubuntu)
 - Windows Server administration
 - Active Directory architecture (Forest, Domain, GPO)
 - DNS and directory services integration
 - Microsoft Certificate Authority (PKI)
 - Patch management and system updates
 - Exchange Server and Office 365 integration
 - Infrastructure security and high availability design
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Information Security, Cyber Security, IT Governance, Risk & Compliance

1. Cyber Security Fundamentals

- Cyber security principles and security architecture
- Threat landscape and cyber attack types

- Malware, phishing, ransomware and insider threats
 - Security best practices and defense mechanisms
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2. Security Operations Center (SOC)

- SOC architecture and operations
 - **Security Information and Event Management (SIEM)**
 - Security event monitoring and alert triage
 - Log analysis and threat detection
 - Incident escalation and response workflow
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3. Incident Response & Digital Forensics

- **Incident Response (IR) lifecycle**
 - Digital Forensics and Incident Response (**DFIR**)
 - Threat investigation techniques
 - Evidence collection and analysis
 - Incident documentation and reporting
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4. Vulnerability Management & Penetration Testing

- **Vulnerability Assessment and Penetration Testing (VAPT)**
 - Security scanning tools and techniques
 - Patch management and vulnerability mitigation
 - Security testing of infrastructure and applications
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5. Application Security

- Secure software development lifecycle (**SSDLC**)
 - **Static Application Security Testing (SAST)**
 - **Dynamic Application Security Testing (DAST)**
 - Secure coding practices
 - Application vulnerability management
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6. Endpoint & Network Security

- Endpoint protection systems
 - **Endpoint Detection and Response (EDR)**
 - **Extended Detection and Response (XDR)**
 - Security tools
 - **Data Loss Prevention (DLP)**
 - **Network Access Control (NAC)**
 - **Privileged Identity Management (PIM)**
 - Network security monitoring and protection
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7. IT Risk Management

- IT risk assessment methodologies
- Risk identification, evaluation and mitigation
- Risk control frameworks and governance
- Risk reporting and monitoring

Risk frameworks

- **FAIR risk model**
 - Sensitivity analysis
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8. IT Governance & Compliance

- IT governance principles
- Compliance management and regulatory requirements
- Internal and external audit processes
- Security policies and control frameworks

Major frameworks

- **ISO 27001**
 - **NIST Cybersecurity Framework**
 - **COBIT**
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9. Security Architecture

- Enterprise security architecture design
- Security control implementation

- Secure infrastructure design
 - Security integration across IT systems
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10. Security Monitoring & Threat Detection

- Continuous security monitoring
 - Threat intelligence and analysis
 - Security event correlation
 - Detection of suspicious activities
-

11. Data Protection & Privacy

- Data protection strategies
- Encryption techniques
- Data confidentiality and integrity
- Privacy regulations and compliance

Regulations

- **GDPR**
 - **DPDP (Digital Personal Data Protection Act)**
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12. IT Infrastructure Security

- Security in operating systems
 - Linux / Unix
 - Windows
 - Database security (Oracle, SQL)
 - Network security management
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13. Risk Metrics & Reporting

- **Key Risk Indicators (KRI)**
 - **Key Control Indicators (KCI)**
 - **Key Performance Indicators (KPI)**
 - Security risk dashboards and reporting tools (Power BI)
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14. Audit & Exception Management

- Exception management processes
 - Risk acceptance workflows
 - Security control testing and validation
 - Documentation and audit evidence preparation
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15. Security Governance & Leadership

- Security program management
 - Coordination with CISO and risk management teams
 - Stakeholder communication and reporting
 - Security strategy development and implementation
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Network Administrator / Network Security

1. Computer Networking Fundamentals

- Networking concepts and architectures
- **OSI and TCP/IP models**
- IP addressing and subnetting
- Network protocols and services

Common protocols

- HTTP / HTTPS
 - FTP
 - DNS
 - DHCP
 - SNMP
 - SSH
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2. Routing & Switching

- Routing fundamentals
- **Border Gateway Protocol (BGP)**
- Static and dynamic routing
- VLANs and trunking
- Switching concepts and configurations

Network devices

- Routers
 - Switches
 - Layer 2 and Layer 3 switching
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3. Network Infrastructure Management

- Network architecture design
 - Network infrastructure deployment
 - Network equipment configuration
 - Network performance optimization
 - Network scalability and capacity planning
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4. Data Center Networking

- Data center network architecture
 - Core, distribution and access layer design
 - Network redundancy and high availability
 - Data center network monitoring
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5. Network Security

- Network security architecture
- Firewall technologies and configuration
- Secure network communication
- Network access control policies

Security devices

- **Web Application Firewall (WAF)**
- **Intrusion Detection System (IDS)**

- **Intrusion Prevention System (IPS)**
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6. Load Balancing Technologies

- Load balancing concepts
- Application load balancing
- High availability and failover

Load balancing tools

- **Citrix Load Balancer**
 - **Array Networks**
 - **Radware**
-

7. Firewall Administration

- Firewall architecture and security rules
- Firewall configuration and management

Firewall platforms

- **Cisco Firewall**
 - **Palo Alto Firewall**
 - **Checkpoint Firewall**
-

8. Network Monitoring & Logging

- Network monitoring tools and techniques
 - Log collection and analysis
 - Network performance monitoring
 - Incident detection and response
-

9. Network Security Compliance

- Network security policies
 - Compliance with organizational standards
 - Vulnerability assessment and remediation
 - Network system hardening
-

10. Cisco Network Technologies

- Cisco routers and switches configuration
 - Cisco Nexus switches
 - Cisco ACI architecture
 - Network device installation and maintenance
-

11. Network Troubleshooting

- Diagnosing network connectivity issues
 - Packet flow analysis
 - Network latency and performance troubleshooting
 - Coordination with L2/L3 support teams
-

12. Infrastructure Management & Upgrades

- Network equipment installation
 - Firmware and software upgrades
 - Network patching and maintenance
 - Vendor coordination for support and upgrades
-

13. Audit & Compliance

- Network audit procedures
 - Security compliance monitoring
 - Documentation and reporting
 - Participation in internal and external audits
-

Production Support /

1. Production Support & Operations

- Production support concepts and responsibilities

- Monitoring and maintaining enterprise applications
 - Ensuring system availability, stability, and performance
 - Handling production incidents and service disruptions
 - Production environment management (24×7 operations)
-

2. IT Service Management (ITSM)

- **ITIL framework fundamentals**
 - Incident management
 - Problem management
 - Change management
 - Service request management
 - Service level agreements (**SLA**) monitoring
-

3. Incident & Crisis Management

- Incident detection and response
 - **Major incident management**
 - Escalation procedures
 - Root Cause Analysis (**RCA**)
 - Preventive and corrective actions
-

4. Application Monitoring & Observability

Monitoring tools

- **Splunk**
- **AppDynamics**
- **Dynatrace**
- **ELK Stack**
- **ServiceNow**
- **Heal**

Topics

- Log analysis and monitoring
- Application performance monitoring

- Alerting and incident detection
-

5. Banking Application Systems

- Core Banking Systems (**CBS**)
- Loan Origination Systems (**LOS**)
- Mobile banking applications
- Enterprise Service Bus (**ESB**)
- API Gateway architecture

Topics

- Enterprise application integration
 - Digital banking systems monitoring
-

6. Middleware & Enterprise Platforms

Middleware technologies

- **WebLogic**
- **WebSphere**
- **IBM MQ**

Concepts

- Middleware architecture
 - Message queuing systems
 - Enterprise application integration
-

7. Database Fundamentals

- Database concepts and administration basics
- Databases used in enterprise systems
 - **Oracle**
 - **SQL**
 - **DB2**

Topics

- Query execution and monitoring
- Database troubleshooting

8. Operational Resilience & Continuity

- **Business Continuity Planning (BCP)**
- **Disaster Recovery (DR) strategies**
- Resilience drills and testing
- High availability architecture

9. Cybersecurity & Compliance

- Cybersecurity basics in production environments
- Fraud monitoring in digital banking systems
- IT risk management frameworks
- Compliance with **RBI guidelines**

10. Ticket Management & Troubleshooting

- Ticket triaging and prioritization
- Following Standard Operating Procedures (**SOPs**)
- Troubleshooting application issues
- Coordination with **L1 / L2 / L3 support teams**

11. Automation & Operational Efficiency

- Automation of repetitive operational tasks
- Monitoring automation scripts
- Operational dashboards and reporting

12. Vendor & Stakeholder Management

- Vendor coordination for issue resolution
- Cross-team collaboration with IT and business teams
- Communication during major incidents
- Stakeholder reporting and service management

13. Reporting & Operational Metrics

- Incident reports and dashboards
 - Service performance metrics
 - Compliance and operational risk reporting
-

Enterprise / Integration / Public Cloud Architect

1. Enterprise Architecture Fundamentals

- Enterprise architecture principles
 - Architecture layers
 - Business architecture
 - Application architecture
 - Data architecture
 - Technology architecture
 - Architecture governance and standards
 - Enterprise architecture frameworks
 - **TOGAF**
-

2. Banking Enterprise Systems

- Architecture of large banking platforms
 - **Core Banking System (CBS)**
 - Digital banking platforms
 - Loan Origination System (**LOS**)
 - Payment platforms

Topics

- System integration and architecture design
 - Platform modernization strategies
-

3. Cloud Computing & Public Cloud Platforms

- Cloud architecture and service models
 - **IaaS, PaaS, SaaS**

Public cloud platforms

- **AWS**
- **Microsoft Azure**
- **Google Cloud Platform**
- **IBM Cloud**
- **Oracle Cloud**

Topics

- Multi-cloud environments
- Cloud-native application architecture
- Cloud migration strategies

4. Cloud Infrastructure & Services

- Compute services
- Storage systems and object storage
- Cloud networking and security
- Event streaming services
- Serverless computing

5. Cloud Migration & Modernization

- Migration of on-premises systems to cloud
- Application refactoring and re-architecture
- Hybrid cloud architecture
- Cloud transformation strategies

6. Microservices & API Architecture

- **Microservices architecture**
- RESTful API development
- API management and integration
- Event-driven architecture

Integration tools

- **Kafka**
 - API gateways
-

7. Containerization & Cloud Platforms

- **Docker containerization**
- **Kubernetes orchestration**
- Cloud container platforms
 - **OpenShift**
 - **Cloud Foundry**

Topics

- Container-based deployments
 - Scalable application architecture
-

8. DevOps & DevSecOps

- DevOps practices and CI/CD pipelines
- Infrastructure automation

Infrastructure as Code (IaC)

- **Terraform**
- **CloudFormation**
- **Ansible**

Topics

- Automated deployments
 - Secure DevOps pipelines
-

9. Application Development Technologies

Frontend technologies

- **React**
- **Angular**
- **React Native**
- **Flutter**

- HTML / CSS / JavaScript

Backend technologies

- **Java / Spring Boot**
- **Node.js**
- **.NET Core**
- **Python**

Data exchange formats

- JSON
 - XML
-

10. Data Architecture & Integration

- Data architecture design
- Data integration and transformation
- ETL pipelines
- Data governance and management

Topics

- Data ingress and egress
 - Distributed data systems
-

11. Infrastructure & Networking

- Network architecture for enterprise systems
 - Internet-facing platform architecture
 - Load balancing and gateway devices
 - Firewalls and network security
-

12. Cloud Security & Governance

- Cloud security best practices
 - Security groups and network security policies
 - Data security and encryption
 - Regulatory compliance in cloud environments
-

13. Monitoring & Observability

- Monitoring tools and observability frameworks
 - Application performance monitoring
 - Logging and alerting systems
-

14. Cloud Operations & Cost Management

- **CloudOps** (cloud provisioning and operations)
 - **FinOps** (cloud cost monitoring and optimization)
 - Cloud resource management
-

15. Architecture Governance & Documentation

- Architecture reviews and standards compliance
 - Risk identification and impact analysis
 - Architecture documentation and blueprints
 - Technology governance processes
-

DevSecOps Engineer

1. DevOps & DevSecOps Fundamentals

- DevOps principles and practices
 - **DevSecOps concept and secure development lifecycle**
 - Software Development Life Cycle (**SDLC**)
 - Agile development methodology
 - Continuous integration and continuous deployment (**CI/CD**)
-

2. Version Control Systems

- **Git fundamentals**
- Git repository management

- Branching and merging strategies
- Release management workflows

Tools

- **GitLab**
 - **Bitbucket**
-

3. CI/CD Pipeline Development

- Continuous integration pipelines
- Automated build and deployment pipelines
- CI/CD pipeline design and implementation

Tools

- **GitLab CI**
 - **Jenkins**
 - **CircleCI**
 - **ArgoCD**
-

4. Containerization & Orchestration

- **Docker containerization**
- Container image management
- Container registry tools
 - **GitLab Registry**
 - **Quay**
 - **JFrog Artifactory**

Container orchestration

- **Kubernetes**
 - **OpenShift Container Platform**
-

5. Linux System Administration

- **Red Hat Enterprise Linux (RHEL) administration**
- Linux commands and utilities
- Package management and patching

- System monitoring and troubleshooting
-

6. Scripting & Automation

- Shell scripting (**Bash**)
- Automation scripting languages
 - **Python**
 - **Perl**
 - **PowerShell**
 - **JavaScript**

Topics

- Task automation
 - Infrastructure automation
 - Build and deployment automation
-

7. Configuration Management

Configuration management tools

- **Ansible**
- **Chef**
- **Puppet**
- **SaltStack**

Topics

- Infrastructure configuration automation
 - Environment configuration management
-

8. Infrastructure as Code (IaC)

- Infrastructure provisioning using code

Tools

- **Terraform**
- **CloudFormation**

Topics

- Automated infrastructure deployment

- Infrastructure version control
-

9. Security in DevSecOps

- Secure software development lifecycle (SSDLC)
- Application security testing

Security tools

- **Static Application Security Testing (SAST)**
- **Dynamic Application Security Testing (DAST)**

Topics

- Vulnerability detection in code
 - Secure application deployment
-

10. Cloud & Network Security

- Cloud security principles
 - Network security in DevOps environments
 - Secure infrastructure configurations
 - Security best practices in CI/CD pipelines
-

11. Monitoring & Logging

- System monitoring and observability
 - Log management and analysis
 - Performance monitoring for applications and infrastructure
-

12. Backup, Recovery & Disaster Management

- Backup and restoration of DevOps infrastructure
- Disaster recovery strategies
- Failover and switchover mechanisms

Example

- **GitLab Geo replication and DR drills**
-

13. DevOps Collaboration Tools

- Project management and collaboration tools
 - **JIRA**
 - **Confluence**

Topics

- Issue tracking
- Development documentation

Web Server / WebLogic Administrator

1. Web Server Fundamentals

- Web server architecture and working principles
- HTTP / HTTPS protocol
- Client-server communication model
- Web hosting and server configuration

Web servers

- **Apache HTTP Server**
- **Nginx**
- **Microsoft IIS**
- **httpd**

2. Application Server Administration

- Application server architecture
- Deployment and management of enterprise applications
- **Oracle WebLogic Server administration**
- Domain configuration and server instances
- Application deployment and version management

3. Proxy & Reverse Proxy Servers

- Proxy server architecture
- **Reverse proxy configuration**
- Load balancing using reverse proxies
- Traffic routing and request forwarding

Tools

- **Nginx**
 - **Apache**
-

4. Application Deployment & Version Management

- Web application deployment strategies
 - Application version control and release management
 - Configuration of application environments
 - Continuous deployment practices
-

5. Server Performance Monitoring

- Monitoring server uptime and performance
 - Resource utilization analysis (CPU, memory, disk)
 - Performance optimization techniques
 - Load balancing concepts
-

6. Security in Web Servers

- **SSL / TLS encryption**
 - Secure communication protocols
 - Authentication and authorization mechanisms
 - Web server hardening and security patches
-

7. Backup & Disaster Recovery

- Backup strategies for applications and databases
- Disaster Recovery (**DR**) planning
- Business continuity strategies
- Data restoration and failover mechanisms

8. Server Maintenance & Troubleshooting

- Log monitoring and analysis
- Disk cleanup and log rotation
- Diagnosing server issues
- Troubleshooting web server errors

9. Automation & Scripting

- **Shell scripting**
- **Python scripting**
- Automation of server deployment and monitoring tasks

Topics

- Script-based monitoring
- Automated deployment scripts

10. Configuration Management Tools

- Infrastructure automation tools
 - **Ansible**
 - **Puppet**

Topics

- Automated configuration management
- Infrastructure provisioning

11. IT Operations & Support

- Ticket management and issue resolution
- Following Standard Operating Procedures (**SOPs**)
- Coordination with **L1 / L2 / L3 support teams**

12. Infrastructure & Resource Management

- Server resource allocation and management
- Server domain configuration

- Infrastructure scalability and high availability
-

Data Architect

1. Data Architecture Fundamentals

- Data architecture principles and frameworks
 - Enterprise data architecture design
 - **Lakehouse architecture (data lake + data warehouse)**
 - Separation of storage and compute layers
 - Data platform governance and architecture standards
-

2. Data Modeling Techniques

- **Enterprise data modeling**
- **Dimensional data modeling**
 - Star schema
 - Snowflake schema

Modeling approaches

- **Inmon model**
- **Kimball methodology**
- **Domain Driven Design (DDD)**

Topics

- Data schema design
 - Conformed dimensions and reference data
 - Data lifecycle management
-

3. Data Lakehouse Technologies

- Lakehouse storage formats
 - **Delta Lake**
 - **Apache Iceberg**

- **Apache Hudi**

Topics

- Data versioning and schema evolution
 - Data partitioning and optimization
 - Large-scale analytics architecture
-

4. Big Data Processing Platforms

- **Apache Spark architecture**
 - Distributed data processing
 - Performance optimization in Spark
 - Integration with relational databases
-

5. Data Integration & Streaming

- Data ingestion and integration pipelines
- Change Data Capture (CDC) patterns

Streaming tools

- **Apache Kafka**
- **Apache Flink**
- **Debezium**
- **Oracle GoldenGate**

Topics

- Event-driven data architecture
 - Real-time data pipelines
-

6. Data Formats & Interoperability

- Data serialization formats
 - **Avro**
 - **Parquet**
 - JSON

Topics

- **OpenAPI standards**

- API-based data integration
 - Data sharing contracts and schema versioning
-

7. Metadata, Lineage & Data Catalog

- Metadata management concepts
- Data lineage tracking
- Data discovery and catalog systems

Tools

- **Apache Atlas**
- **Collibra**

Topics

- Semantic layers and data discoverability
-

8. Data Security Architecture

- Data security in big data systems
- Encryption and key management
- Data masking and tokenization
- Access control models
 - **RBAC (Role Based Access Control)**
 - **ABAC (Attribute Based Access Control)**

Security tools

- **KMS / HSM**
 - **Apache Ranger**
-

9. Data Governance & Quality

- Data governance frameworks
- Data quality metrics
 - Data freshness
 - Data completeness
 - Data accuracy

Topics

- Data observability
 - Data stewardship and governance policies
-

10. Disaster Recovery & Data Resilience

- Disaster recovery planning
- **RPO (Recovery Point Objective)**
- **RTO (Recovery Time Objective)**

Topics

- Data replication and backup strategies
 - High availability data architecture
-

11. Cloud & Hybrid Data Platforms

- Hybrid cloud data architecture
- Data platforms on public cloud environments
- Integration between on-premise and cloud systems

Topics

- **MeitY empanelled cloud environments**
 - Cloud data services and infrastructure
-

12. Big Data Cost & Performance Optimization

- **FinOps for big data platforms**
 - Cost optimization strategies
 - Workload management and resource allocation
 - Storage and compute efficiency
-

13. GenAI Data Infrastructure

- Data architecture for **Generative AI systems**
 - **Vector databases** and embeddings
 - AI data pipelines and model data preparation
-

14. Architecture Governance & Documentation

- High Level Design (**HLD**) and Low Level Design (**LLD**)
- Architecture Review Board (**ARB**) governance
- Data architecture documentation and standards
- Compliance and audit readiness

Data Engineer / Data Quality / ETL / PLSQL / BI

1. Data Engineering Fundamentals

- Data engineering architecture and workflows
- Data ingestion and integration pipelines
- Batch processing and **ELT / ETL architectures**
- Handling large-scale enterprise data platforms

2. SQL & Database Programming

- **SQL query writing**
- **PL/SQL / T-SQL programming**
- Stored procedures and functions
- Query optimization techniques
- Data extraction and transformation

3. Distributed Data Processing

- **Apache Spark architecture**
- Distributed data processing concepts
- Spark batch and streaming jobs
- Spark performance optimization

Topics

- Partitioning
- Bucketing
- Memory and executor configuration

4. Data Ingestion & Integration

- Data ingestion pipelines from enterprise systems
 - Core Banking Systems (CBS)
 - Loan Origination Systems (LOS)
 - Card platforms

Topics

- Handling late-arriving data
- Replay and rerun pipeline strategies

5. Streaming Data Processing

- Real-time data streaming architectures

Streaming technologies

- **Apache Kafka**
- **Apache Flink**
- **Apache NiFi**

Topics

- Event-driven data pipelines
- Stream processing design

6. Change Data Capture (CDC)

- CDC architecture and patterns
- Real-time data replication

Tools

- **Debezium**
- **Oracle GoldenGate**

Topics

- Data synchronization across systems

7. Data Formats & Storage

Common data formats

- **Parquet**
- **Avro**
- **JSON**

Topics

- Data serialization and compression
 - Data storage optimization
-

8. ETL / ELT Tools

Enterprise ETL tools

- **Informatica**
- **Talend**
- **Pentaho**
- **IBM DataStage**
- **Apache NiFi**

Topics

- Data transformation workflows
 - Data pipeline orchestration
-

9. Workflow Orchestration

- Data pipeline orchestration
- Job scheduling and dependency management

Tools

- **Apache Airflow**

Topics

- Workflow automation
 - Job monitoring and alerts
-

10. Data Quality & Governance

- Data profiling and validation
- Data quality rules and monitoring
- Data lineage tracking

Metrics

- Data completeness
 - Data accuracy
 - Data freshness
-

11. API Development for Data Services

- RESTful API development for data access
- Data product APIs and low-latency data services

Technologies

- **Java / Spring Boot**
- **.NET**

Security topics

- **OAuth2 authentication**
 - API rate limiting
 - Idempotency
-

12. CI/CD for Data Pipelines

- CI/CD pipelines for data engineering workflows
- Automated testing and deployment

Tools

- **Git**
- **Jenkins**
- **GitLab**

Topics

- Git workflows and version control
 - Artifact repositories
-

13. Monitoring & Observability

- Monitoring data pipelines and jobs
- Logging and tracing in distributed systems
- Performance dashboards and alerts

Topics

- Metrics and observability frameworks
-

14. Security & Access Control

- Secure data engineering practices
- Secrets management and access control
- IAM and network security

Topics

- Secure data pipelines
 - Restricted network environments
-

15. AI & Advanced Data Engineering

- Use of **Generative AI and NLP for data engineering**
 - Automated data pipeline development
 - AI-enabled data processing frameworks
-

Data Scientist

1. Data Science Fundamentals

- Data science lifecycle and workflow
 - Data collection, cleaning, and preprocessing
 - Exploratory Data Analysis (**EDA**)
 - Feature engineering and feature selection
-

2. Programming for Data Science

Languages

- **Python**

Libraries

- **Pandas**

- **NumPy**
- **Scikit-learn**

Topics

- Data manipulation and transformation
 - Data visualization and analysis
-

3. Machine Learning Algorithms

Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Gradient Boosting (**XGBoost / LightGBM**)

Unsupervised Learning

- K-Means clustering
 - Hierarchical clustering
 - Dimensionality reduction (PCA)
-

4. Model Evaluation & Experimentation

- Model validation techniques
- Cross validation methods
- Evaluation metrics
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - ROC-AUC

Topics

- Experiment design
 - Model performance comparison
-

5. Time Series & Survival Analysis

Time series topics

- Trend and seasonality
- Forecasting models
- ARIMA / SARIMA basics

Survival analysis

- Hazard functions
 - Risk modeling for collections and churn prediction
-

6. Explainable AI (XAI)

Model interpretability techniques

Tools

- **SHAP (Shapley Additive Explanations)**
- **LIME (Local Interpretable Model-Agnostic Explanations)**

Topics

- Feature importance analysis
 - Model transparency and explainability
-

7. Big Data Processing

- Distributed data processing concepts
- **Apache Spark for large-scale data analysis**

Topics

- Data pipelines and distributed ML workflows
-

8. MLOps & Model Lifecycle Management

Model lifecycle processes

- Model training pipelines
- Model validation pipelines
- Model deployment workflows

Tools

- **MLflow**

- **DVC**
- **Apache Airflow**

Topics

- Experiment tracking
 - Model registry and versioning
-

9. Model Deployment

Deployment strategies

- Batch scoring pipelines
- Real-time model inference APIs

Technologies

- **FastAPI**
- **Flask**

Topics

- API-based model serving
 - Containerized deployment
-

10. Model Monitoring

- Model performance monitoring
- **Data drift detection**
- **Concept drift detection**

Topics

- Model retraining strategies
 - Alerting systems
-

11. Data Governance & Security

- Data privacy and protection
- Secure data integration

Techniques

- Data masking
- Tokenization

Topics

- Consent management and regulatory compliance
-

12. Responsible AI & Compliance

- Ethical AI principles
- Bias and fairness detection
- Responsible AI frameworks

Regulations

- **DPDP (Digital Personal Data Protection Act)**
-

13. AI Use Cases in Banking

Important AI applications

- Fraud detection and AML analysis
 - Credit risk modeling
 - Collections propensity modeling
 - Next Best Action (**NBA**)
 - Next Best Offer (**NBO**)
 - Operations automation
-

14. Data Integration & APIs

- Integration of internal and external data sources
 - API-based data services
 - Data contracts and schema management
-

15. BI & Business Impact Measurement

- Integration with Business Intelligence systems
 - Measuring model performance in production
 - Continuous improvement of AI models
-

1. Generative AI Fundamentals

- Concepts of **Generative Artificial Intelligence**
- Architecture of **Large Language Models (LLMs)**
- Transformer architecture and attention mechanism
- Pretrained vs fine-tuned models

Examples of LLMs

- **GPT models**
- **BERT**
- Similar transformer-based models

2. Large Language Models (LLM)

- LLM architecture and training workflow
- Tokenization and embeddings
- Pretraining and fine-tuning methods
- Prompt-based inference

Topics

- Model scaling
- Transfer learning

3. Prompt Engineering

- Prompt design techniques
- Context engineering for LLMs
- Prompt optimization strategies
- Few-shot and zero-shot prompting

4. Retrieval Augmented Generation (RAG)

- **RAG architecture and workflow**
- Integration of LLMs with knowledge bases
- Retrieval pipelines for contextual generation

- Data indexing and document retrieval

Topics

- RAG pipelines
 - Context-aware response generation
-

5. Embeddings & Vector Databases

- Text embedding models
- Similarity search techniques
- Semantic search systems

Vector databases

- Vector indexing and retrieval
 - Embedding storage and retrieval workflows
-

6. Agentic AI & Multi-Agent Systems

- **Agent-based AI systems**
 - Multi-agent AI architectures
 - Autonomous AI workflows
 - Task orchestration using AI agents
-

7. AI Model Development Frameworks

Deep learning frameworks

- **TensorFlow**
- **PyTorch**
- **Hugging Face Transformers**

Topics

- Model training and fine-tuning
 - Model evaluation and benchmarking
-

8. Data Pipelines for AI

- Data ingestion and processing pipelines
- Handling text and document datasets

- Data cleaning and preprocessing for NLP
 - Data pipeline architecture for AI systems
-

9. Model Deployment & Inference

- LLM deployment architectures
- Model serving frameworks
- Optimization techniques for inference
- Low-latency AI inference pipelines

Topics

- Efficient memory usage
 - Model compression and quantization
-

10. GPU & Distributed Computing

- **GPU architecture for deep learning**
 - GPU clusters and parallel computing
 - Distributed model training
 - Hardware optimization for AI workloads
-

11. Cloud AI Infrastructure

Cloud platforms

- **AWS**
- **Microsoft Azure**
- **Google Cloud Platform**

Topics

- AI workloads in cloud environments
 - Infrastructure scaling for AI systems
 - Cloud-based AI services
-

12. Containerization & AI Infrastructure

- **Docker containerization**
- **Kubernetes orchestration**

Topics

- Scalable AI deployment
 - Container-based model serving
-

13. AI Lifecycle Management

- End-to-end AI lifecycle
 - Model training
 - Validation
 - Deployment
 - Monitoring

Topics

- Model versioning
 - Performance monitoring
-

14. AI Security & Governance

- AI safety and guardrails
 - Data privacy in AI systems
 - Ethical AI and responsible AI practices
-

15. Cost Optimization & Infrastructure Management

- Cloud resource monitoring
- Cost optimization strategies for AI workloads
- Infrastructure rightsizing and scaling

Topics

- Resource tagging and monitoring
- Infrastructure efficiency

IT Officers

1. IT Infrastructure Management

- IT infrastructure components
 - Hardware systems
 - Software platforms
 - Networking systems
 - IT system installation, configuration, and maintenance
 - Infrastructure monitoring and troubleshooting
-

2. Banking IT Systems

- **Core Banking System (CBS) architecture**
- Digital banking systems
 - ATM systems
 - Mobile banking
 - Internet banking
 - UPI systems
 - Kiosk banking

Topics

- Banking application integration
 - Digital payment infrastructure
-

3. Operating Systems Administration

- Windows operating system administration
 - Linux operating system basics
 - System configuration and management
 - Server monitoring and troubleshooting
-

4. Networking Fundamentals

- Computer networking concepts
- IP addressing and subnetting
- Network protocols

Common protocols

- HTTP / HTTPS
- FTP
- DNS
- DHCP

Topics

- Network troubleshooting
 - Network security basics
-

5. Programming & Application Development

Programming languages

- **Python**
- **NodeJS**
- **PHP**
- **Java / Spring Boot**
- **Go**
- **Ruby**
- **.NET / C#**

Topics

- Object-Oriented Programming (**OOP**) concepts
 - Application development basics
-

6. Database Management

Relational databases

- **Oracle**
- **PostgreSQL**
- **MySQL**

NoSQL databases

- **MongoDB**

Topics

- SQL queries
- **PL/SQL programming**

- Database troubleshooting
-

7. Application Monitoring & Troubleshooting

- Application monitoring techniques
 - Log analysis and troubleshooting
 - Audit trail monitoring
 - Performance monitoring
-

8. IT Service Management

- IT service support processes
 - Ticket management and issue resolution
 - Coordination with **L1 / L2 / L3 support teams**
 - Standard Operating Procedures (**SOPs**)
-

9. Email Server Administration

Email platforms

- **Microsoft Exchange Server**
- **Office 365**
- **IBM Notes / HCL Notes**

Email protocols

- **SMTP**
- **POP**
- **IMAP**

Topics

- Email routing and server monitoring
 - Email account management
 - Distribution lists and policies
-

10. Email Security

- Spam filtering systems
- Email encryption techniques

- Antivirus protection for email servers
 - Secure messaging infrastructure
-

11. Data Backup & Disaster Recovery

- Backup strategies and policies
 - Data recovery processes
 - Disaster Recovery (**DR**) planning
 - Business continuity management
-

12. IT Project & Vendor Management

- IT project management fundamentals
 - Vendor management and coordination
 - Technology evaluation and implementation
-

13. IT Documentation & Reporting

- IT documentation practices
 - System configuration documentation
 - **MIS reporting and monitoring dashboards**
-

14. Technical Support & User Training

- End-user support and troubleshooting
 - Helpdesk operations
 - User training and support documentation
-

15. IT Security Basics

- Information security principles
- Secure system configuration
- IT policy implementation
- Basic cybersecurity practices

1. Risk Management Fundamentals

- Concepts of **risk in banking and financial institutions**
- Types of financial risks
 - Credit Risk
 - Market Risk
 - Operational Risk
 - Liquidity Risk
- Risk identification, measurement, and mitigation techniques

2. Credit Risk Management

- Credit risk assessment frameworks
- Loan evaluation and credit appraisal
- Credit scoring models
- Non-Performing Assets (**NPA**) management
- Credit risk mitigation techniques

Topics

- Probability of Default (**PD**)
- Loss Given Default (**LGD**)
- Exposure at Default (**EAD**)

3. Market Risk Management

- Market risk concepts and measurement
- Interest rate risk and exchange rate risk
- Equity and commodity risks

Key models

- **Value at Risk (VaR)**
 - Stress testing
 - Scenario analysis
-

4. Liquidity Risk & Asset Liability Management

- **Asset Liability Management (ALM)**
 - Liquidity risk measurement
 - Gap analysis and duration analysis
 - Liquidity coverage ratio (LCR)
-

5. Operational Risk Management

- Operational risk frameworks
- Risk event identification
- Loss event management
- Internal controls and risk mitigation

Topics

- Business continuity planning (**BCP**)
 - Disaster recovery planning (**DR**)
-

6. Banking Regulations & Compliance

Regulatory frameworks

- **Basel II / Basel III guidelines**
- **RBI risk management guidelines**
- Banking regulatory compliance

Topics

- Capital adequacy norms
 - Risk-weighted assets (**RWA**)
-

7. Treasury & Financial Markets

- Treasury operations in banks
- Fixed income securities

- Interest rate derivatives
- Foreign exchange risk management

Topics

- Yield curve analysis
 - Duration and convexity
-

8. Quantitative Risk Analysis

- Statistical methods in risk analysis
- Probability distributions
- Regression analysis
- Time series analysis

Tools

- **SPSS**
 - **SAS**
-

9. Financial Modeling

- Financial forecasting models
- Risk modeling techniques
- Stress testing and scenario modeling

Topics

- Portfolio risk management
 - Sensitivity analysis
-

10. Data Analytics for Risk Management

- Data-driven risk assessment
- Predictive analytics in credit risk
- Risk dashboards and reporting

Tools

- Statistical software (SPSS / SAS)
-

11. Risk Governance & Reporting

- Risk governance frameworks
- Risk policies and procedures
- Risk reporting to senior management

Metrics

- **Key Risk Indicators (KRI)**
 - Risk appetite framework
-

12. Banking Products & Risk Exposure

- Retail banking products
- Corporate banking products
- Investment products

Topics

- Risk exposure across financial products
-

13. Risk Audit & Internal Controls

- Internal audit and risk control systems
 - Compliance monitoring
 - Risk review and reporting
-

14. Portfolio Risk Management

- Diversification strategies
- Portfolio risk measurement
- Risk-adjusted return metrics

Examples

- **Sharpe Ratio**
 - **RAROC**
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15. Financial Risk Certifications Knowledge

Important frameworks covered in professional certifications:

- **FRM (Financial Risk Manager)**
- **CFA (Chartered Financial Analyst)**

- **PRM (Professional Risk Manager)**
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Good IT Officer